Tejas Rao | *Aspiring Researcher*

Email: tejas_rao.sias23@krea.ac.in | LinkedIn: linkedin.com/in/tejasnrao | Web: tejas-n-rao.github.io/about

Education

Krea University, B.Sc in Biological Sciences and Data Science

- **Grade**: Junior (3rd Year) | GPA: **9.79**/10 (On Dean's List since Freshman year)
- **Thesis** (till April, 2027): Deep Learning for Image Registration tasks on MRI data | Under Researchers from **Harvard Medical School** and **Massachusetts General Hospital**
- **Coursework:** *Data Science:* Applied Statistics, Python for Data Science, Introduction to Statistical Learning, Responsible AI, Applied Dynamical Systems, Data Analytics, Data Structures and Algorithms, Geometry for Computer Vision, Linear Algebra, Introduction to Calculus, Discrete Mathematics
  Biological Sciences: Biostatistics (R), Biochemistry, Microbial Genomics, Genetics & Advanced Cell Biology Lab, Anatomy & Physiology, Physics for Chemists and Biologists, Evolution, Ecology & Conservation Biology, Life at Different Scales
  Other: Scientific Reasoning, Written And Oral Communication (Academic Writing), Design Thinking, Ethics

Digital SAT: Score of **1500**/1600 (98^{th} Percentile)

- Mathematics Section: **800**/800 (99^{th} Percentile) | Reading Section: **700**/800 (93^{rd} Percentile)

Grade 12 (CBSE - Sciences), NPS Agara: Grade of **96.87%** (99^{th} Percentile)

Publications

Investigating Health Literacy and Sociodemographic Factors in College Students Jul 2025
Shyam Kumar Sudhakar, Divij Doshi, Gayatri Nair, ***Tejas Rao*** | *Scientific Reports (Nature Portfolio)*

Interpretable Deep Learning Predicts Polycythemia Vera and Identifies Novel Genomic Associations
Tejas Rao†, Chepsy C. Philip, Bani Jolly, Aakash Chozakade, Minu Luckose, Bobby George, Sudhir Venkatesh, Bonnie George, Anupa Jacob, Vinod Scaria | *Accepted to American Society of Haematology's 2025 Annual Meeting (†: Presenting Author)*

Depolarization Block Susceptibility of Inhibitory Fast-Spiking Neurons: Biophysical Mechanisms that can Contribute to Seizure Initiation and Propagation
Shyam Kumar Sudhakar, ***Tejas Rao***, Omar J Ahmed | *In Preparation, to be submitted to Epilepsia (IF: 6.6)*

From N- to O-Glycoproteomics: Revisiting Astral Narrow-Window DIA Data for O-Glycopeptide Discovery
Kathirvel Alagesan*, ***Tejas Rao**** | *In Preparation (∗: Equal Contribution)*

Investigating the Association Between Skin Health and Stress in University Students
Shyam Kumar Sudhakar, Sunita Makhijani, ***Tejas Rao***, Gautam Bannerjee | *In preparation (Journal Submission)*

Experience

LINC Fellow, Harvard + MGH

- Selected as an undergraduate fellow embedded within the computational team at the Centre for Large-Scale Imaging of Neural Circuits (PI: Dr. Anastasia Yendiki), an NIH BRAIN CONNECTS project spanning Harvard, MIT, and MGH.
- Conducting Bachelor's Thesis under Dr. Ricardo Gonzalez (Harvard, MGH) on using domain randomisation and deep learning in neuroimaging (MRI) tasks to improve robustness and generalisation.
- Working with large-scale neuroimaging datasets and HPC workflows to prototype, train, and evaluate deep learning pipelines in Python/PyTorch.

Bioinformatics Intern, Karkinos Healthcare

- Supervised by Dr. Bani Jolly and Dr. Vinod Scaria in the Bioinformatics R&D Division of a precision oncology company.
- Collaborating with AI/ML and Bioinformatics teams to characterise differences in Whole Exome Sequencing (WES) data between healthy individuals and cancer patients, with a focus on polygenic malignancies.
- Developed a novel, explainable deep learning methodology to predict the presence of polygenic malignancies

(e.g., Polycythemia Vera) with high fidelity, providing gene-level importance scores that recover known and highlight putative novel genomic associations.
- Implemented reproducible Python-based pipelines (HPC job scheduling, VCF handling, data preprocessing, model training) and contributed to results that were accepted to ASH 2025 and are being prepared for journal publication.

Research Collaborator, Max Planck Unit for the Science of Pathogens – Remote
- Collaborating with the Proteomics Research Platform (incubated in Charpentier Lab, Nobel – 2020) under Dr. Kathirvel Alagesan on quantitative bottom-up mass spectrometry for O-glycoproteomics.
- Analysing large-scale DIA datasets to evaluate methods for profiling the O-glycoproteome; responsibilities include data cleaning, visualisation, and interpretation of glycopeptide-level trends.
- Contributing to manuscript writing and figure generation, with a focus on communicating complex proteomics workflows to interdisciplinary audiences.

Work-Study Student, Krea University
- Co-authoring a manuscript with Prof. Shyam Sudhakar & Prof. Omar Ahmed (UMichigan Ann-Arbor) on biophysical mechanisms underlying depolarisation block in fast-spiking (FS) neurons and their role in epileptic cell states.
- Performed data analysis and visualisation on simulations from biophysically accurate in-silico neuron models, focusing on parameter sensitivity and network-level behaviour.
- Conducted exploratory analysis and hypothesis testing on a pilot study exploring biological, neurological and sociodemographic determinants of mental health in a university cohort.

Teaching Assistant, Krea University
- Teaching Assistant for "Python for Data Science" (DATA201) for a cohort of 110+ students.
- Conducted tutorials and problem-solving sessions, clarified concepts in Python, data wrangling and introductory machine learning, and supported students with debugging and project development.
- Helped design assignments and assessments, graded submissions, and maintained consistent, constructive feedback to support student learning outcomes.

Research Intern and Trainee, Indian Institute of Technology, Delhi
- Conducted a detailed literature review on Deep Learning and Quantum ML in Drug Discovery, synthesising insights from 120+ publications into structured notes and summaries.
- Independently explored Quantum ML models (QGANs & VQE) for small-molecule generation, focusing on how training strategies affect mode collapse and chemical diversity.
- Completed a one-month certification on Quantum Computing & ML (IIT-D: QCML), gaining foundational understanding of quantum circuits, variational algorithms and their applications.

Projects

Deep Learning on EEG Data for Haemoglobin Level Classification
- Working with Prof. Shyam Sudhakar to build deep learning models that classify haemoglobin levels from tensorised EEG representations in an Indian cohort.
- Designing an end-to-end pipeline including preprocessing, feature extraction, model training and evaluation using graph-based and other neural architectures in Python.
- Comparing baseline models against more expressive architectures to understand which aspects of EEG activity most strongly correlate with haemoglobin-related variation.

Differential Effect of Gap Junctions on Depolarisation Block
- Investigated how gap junction coupling can both promote and prevent the propagation of depolarisation block, an epileptic cell state, in networks of neurons.
- Collaborated with Prof. Shyam Sudhakar to design in-silico experiments probing network-specific parameters (e.g., connectivity patterns, coupling strength) using biophysically realistic neuron models.
- Implemented and analysed simulation batches on HPC resources, extracting metrics that motivate future experimental and grant-funded work (anticipated continuation in October 2025).

Conversational RAG Chatbot for University Syllabus Queries

- Led requirements gathering (functional and non-functional) to scope a chatbot that answers curriculum-related queries for university students.
- Co-designed and implemented a Retrieval-Augmented Generation (RAG) chatbot in Python, integrating vector search over syllabus documents with an LLM-based conversational layer to provide contextual answers.
- Collaborated with a team of 3 students to containerise and deploy the system on AWS, performing basic load- and stress-testing to ensure scalability and stability before rollout.

Skillset

Technical Skills:

- *Programming & Data Science:* Python (Numpy, Pandas, Scipy, Matplotlib, Seaborn), R (Tidyverse, ggplot); Experience building ML workflows, implementing deep learning models in PyTorch, performing statistical analysis and hypothesis testing.
- *AI/ML & LLMs:* Sklearn, Explainable AI techniques, Graph Neural Networks, LangChain & LangGraph, RAG pipeline development, model evaluation and optimisation.
- *Bioinformatics & Computational Biology:* Genome analysis (VCF handling, sequence alignment, phylogenetics), variant prioritisation, gene-level interpretation; Tools: NEURON (hoc), RDKit, AlphaFold, basic proteomics workflows including O-glycopeptide analysis.
- *High-Performance Computing:* Bash scripting, cluster job scheduling (PBS), containerisation (Docker), version control (Git), reproducible data pipelines.
- *Cloud & Deployment:* AWS (EC2, S3), Streamlit UI development, deployment with scalability considerations.
- *Wet Lab:* Microbial culture, Drosophila handling & genetics, qPCR, SDS-PAGE, buffer & reagent preparation, microscopy (oil-immersion, stereo), electrophoresis, standard staining techniques.

Additional Skills:

- *Web & Product Skills:* UI/UX design, basic SEO, building content-driven websites (Wix), prototyping interactive tools for users.
- *Media & Content:* Video editing (DaVinci Resolve), photo editing (Photoshop, Lightroom), audio editing (Ableton Live, Audacity), visual asset creation.
- LaTeX for scientific writing; clear communicator with experience explaining complex concepts to audiences in academic and pedagogical environments.

Extracurricular

- Volunteered for the safe rescue and release of over **100** snakes, coordinating with local residents and authorities to ensure safe capture, identification and relocation while building trust with the community.
- Volunteered at ‘Soil and Soul’, a social enterprise offering eco-friendly alternatives to chemical-based products; contributed to the design and development of two websites (including an e-commerce platform), improving discoverability and user experience.
- Interned at BioEnzyme Entrepreneur’s Academy – a women-run non-profit promoting sustainable solutions and training ‘Eco-Entrepreneurs’; helped build the organisation’s website and produced multimedia content (articles, animated videos, banners) to support outreach and education.
- Raised funds to support mid-day meals for nearly **250** underprivileged children for one year (Akshaya Patra Mid-Day Meals Scheme), coordinating with peers and donors to maximise impact.